UNIVERSITY OF SUSSEX

Data Logging and Classification

Where engineering meets AI

Where engineering meets AI

UNIVERSITY OF SUSSEX

Introduction

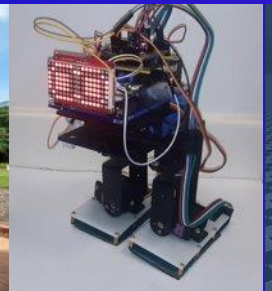
Dexter Shepherd

Robotics researcher at the University of Sussex

Lead for international AI mentoring program

萨塞克斯大学（布莱顿）机器人研究员，

国际人工智能导师项目负责人



UNIVERSITY OF SUSSEX

University of Sussex Overview

About Sussex

- Established in **1961**
- **16,000+** students
- **1st in the world** for Development Studies (QS World University Rankings 2019-25)
- Joint **1st in the world** for research impact in Physics (Shanghai Ranking of Academic Subjects 2025)
- **6 presidents**
- **3 Nobel Prize** winners
- **159+** nationalities represented

Where is the University of Sussex?

Where is the University of Sussex?

Our leafy green campus is situated amongst the beautiful South Downs National Park and just minutes from the vibrant seaside city of Brighton & Hove.

Brighton – the UK’s happiest city

- Brighton has been voted one of the best cities in the UK (Time Out Best Cities in the World 2025), and its residents voted it the happiest UK city 2025
- Visit our famous pebble beach, parks and South Downs for fresh air
- Explore eclectic cafes, international eateries and vintage shops in the vibrant Lanes
- Buzzing culture: home to the UK’s oldest purpose-built cinema, live music, theatres and comedy clubs
- Year-round events: Brighton Festival, Brighton Pride and Brighton and Hove Albion FC
- Easy links: 1 hour to London, 15 mins to Lewes

University structure

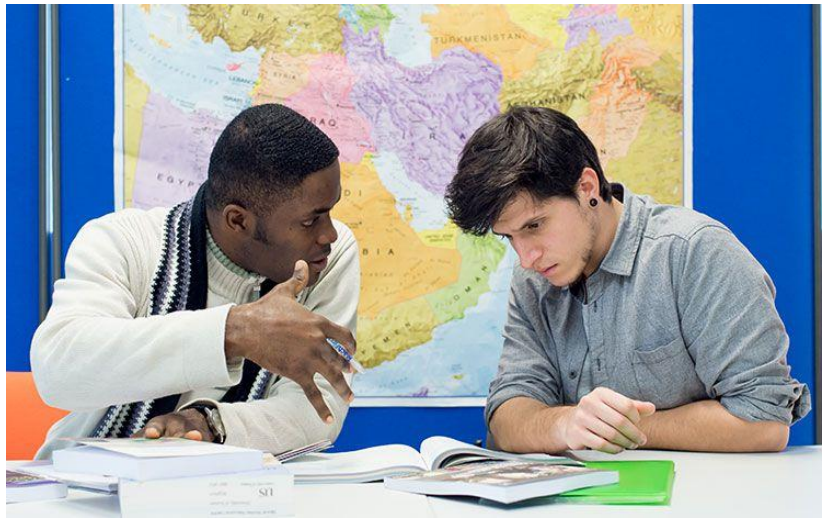
Our four Faculties take an interdisciplinary approach to study and driving forward development in research and teaching.

Each faculty contains a mix of Schools, departments and subject areas.

Media, Arts & Humanities

Science, Engineering & Medicine

Social Sciences



Business School

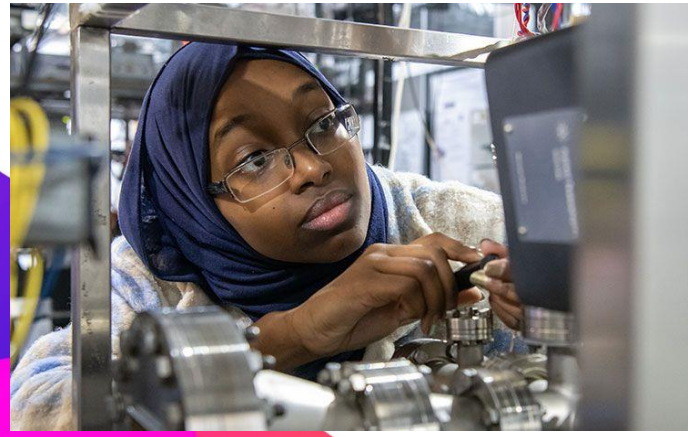
Faculty of Science, Engineering and Medicine

Our Faculty brings together five schools of study and a broad range of disciplinary expertise.

Life Sciences

Psychology

Mathematics and Physical Sciences



Engineering and Informatics

Brighton and Sussex Medical School

Centres of Excellence

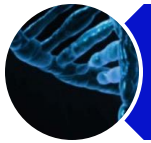
Our centres of excellence are drawing together world-leading experts and innovative approaches, creating a critical mass of knowledge, skills and training – and proving that a challenge is only impossible until it's done.

Sussex AI

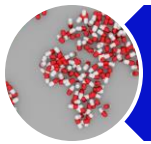
Sussex Neuroscience

Sussex Centre for Quantum Technologies

Sussex Centre for Consciousness Science



Genome Damage and Stability Centre



Centre for Global Health Research

Sussex Sustainability

Our rankings

- **=1st in the world** for research impact in Physics (Shanghai Ranking Global Ranking of Academic Subjects 2025)
- **Top 5 in the UK** for Chemistry (Guardian University Guide 2026)
- **=15th in the UK** for Psychology (QS World University Rankings by Subject 2025)
- **11th in the UK** for Psychology (The Times Higher Education World University Rankings by Subject 2025)
- **Top 20 in the UK** for Mechanical Engineering (The Guardian University Guide 2026)
- **Top 20 in the UK** for Product Design (The Guardian University Guide 2026)
- **100% of our research impact** in Computer Science and Informatics was assessed to be world leading or internationally excellent (REF 2021)
- **Top 10 in the UK** for student satisfaction with the quality of our teaching in Mathematics (Complete University Guide 2026)

Why Sussex

UNIVERSITY OF SUSSEX

•Interdisciplinary teaching and research, encouraging **collaboration across different fields.**
跨学科教学与研究，鼓励跨领域合作。

•**Strong global reputation**, consistently ranked among leading universities worldwide.
享有卓越的国际声誉，长期位居世界顶尖大学行列

•Beautiful green campus located within the **South Downs National Park.**
校园绿意盎然，坐落于南唐斯国家公园之内

•Close **to the vibrant seaside city of Brighton**, known for its **culture, music, and creativity.**
毗邻充满活力的海滨城市布莱顿，这座城市以文化、音乐与创意氛围著称

•Diverse international community with students and staff from **over 100 countries.**
多元包容的国际社区，师生来自全球 100 **多个国家和地区**

•Active student life with **hundreds of societies, sports clubs, and events**.
丰富多元的校园生活，拥有数百个学生社团、体育俱乐部及各类活动。

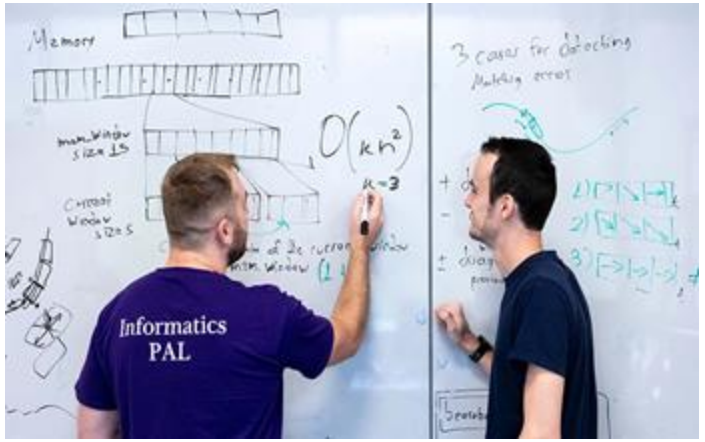
Engineering and Informatics at Sussex

Great environment full of social events and hackathons

充满活力的社交环境，涵盖丰富的校园活动与黑客松大赛

Research led modules, benefiting directly off expert cutting edge work

以研究为主导的课程模块，助你直接汲取专家学者的前沿科研成果



Mentor support schemes to help you if you fall behind

完善的导师支持体系，为你的学业进度与学术挑战保驾护航

Courses in informatics

Undergraduate courses

- Business Analytics BSc
- Business Analytics (with a professional placement year) BSc
- Computer Science BSc
- Computer Science MComp
- Computer Science (with an industrial placement year) BSc
- Computer Science (with an industrial placement year) MComp
- Computer Science and Artificial Intelligence BSc
- Computer Science and Artificial Intelligence (with an industrial placement year) BSc
- Computing Sciences (with a foundation year) BSc
- Computing for Business and Management BSc
- Computing for Business and Management (with an industrial placement year) BSc
- Computing for Digital Media and Games BSc
- Computing for Digital Media and Games (with an industrial placement year) BSc
- Finance and Technology (FinTech) BSc
- Finance and Technology (FinTech) (with a professional placement year) BSc
- Media Production BA
- Media and Communications BA
- Software Engineering BSc

本科课程

- 商业分析理学士学位
- 商业分析（含一年专业实习）理学士学位
- 计算机科学理学士学位
- 计算机科学硕士
- 计算机科学（含一年工业实习）理学士学位
- 计算机科学（含一年工业实习）硕士
- 计算机科学与人工智能理学士学位
- 计算机科学与人工智能（含一年工业实习）理学士学位
- 计算机科学（含预科课程）理学士学位
- 商业与管理计算机科学学士学位
- 商业与管理计算机科学（含一年实习）理学士学位
- 数字媒体与游戏计算理学士学位
- 数字媒体与游戏计算（含一年实习）理学士学位
- 金融与科技（FinTech）理学士学位
- 金融与科技（含专业实习年）理学士学位
- 媒体制作学士学位
- 媒体与传播学士学位
- 软件工程理学士学位

AI at Sussex (Undergraduate)

Year one:

Focus on coding skills and mathematical concepts

Year two:

You have more options to pick modules and branch into areas such as philosophy, Neuroscience, psychology or software development. This year you have core modules for machine learning (computer vision, natural language processing and evolutionary robotics)

Year three:

Final year project and mainly selectable modules from multidisciplinary areas

第一年
专注于编程技能与数学基础

第二年
可选择的课程模块更多，能够分流至哲学、神经科学、心理学或软件开发等方向。

本年设有机器学习核心模块，包括**计算机视觉、自然语言处理、进化机器人学**。

第三年
完成毕业设计，课程以跨学科方向的选修模块为主。

AI at Sussex (Masters)

https://www.sussex.ac.uk/study/masters/courses/advanced-artificial-intelligence-mres

https://www.sussex.ac.uk/study/masters/courses/artificial-intelligence-and-adaptive-systems-msc

Masters courses

- Advanced Artificial Intelligence MRes
- Advanced Computer Science MSc
- Advanced Mobile Communications with AI MSc
- Advanced Mobile Communications with AI (with an industrial placement year) MSc
- Artificial Intelligence and Adaptive Systems MSc
- Artificial Intelligence and Adaptive Systems (with an industrial placement year) MSc

硕士课程

- 高级人工智能硕士研究
- 高级计算机科学硕士
- 人工智能高级移动通信硕士
- 人工智能高级移动通信（含一年工业实习）理学硕士
- 人工智能与自适应系统理学硕士
- 人工智能与自适应系统（含一年工业实习）理学硕士

AI at Sussex (PhD)

A diverse range of research areas in world leading AI research.

Today’s task

Building fitness trackers

How do smart watches work?

Accelerometers monitor acceleration in various directions

Smart watches track this and use machine learning models to interpret this information

From these movements, models can classify what sort of exercise activity is being done.

加速度计可以监测不同方向上的加速度。

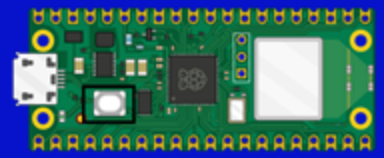
智能手表会跟踪这些数据，并使用机器学习模型来解读这些信息。

通过这些运动数据，模型可以分类判断正在进行的运动类型。

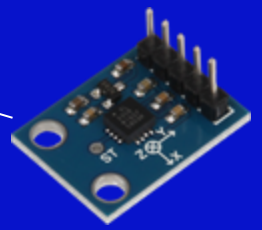
Key Components for fitness

Screen / 屏幕 或 显示屏

Controller / 控制器



Accelerometer / 加速度计

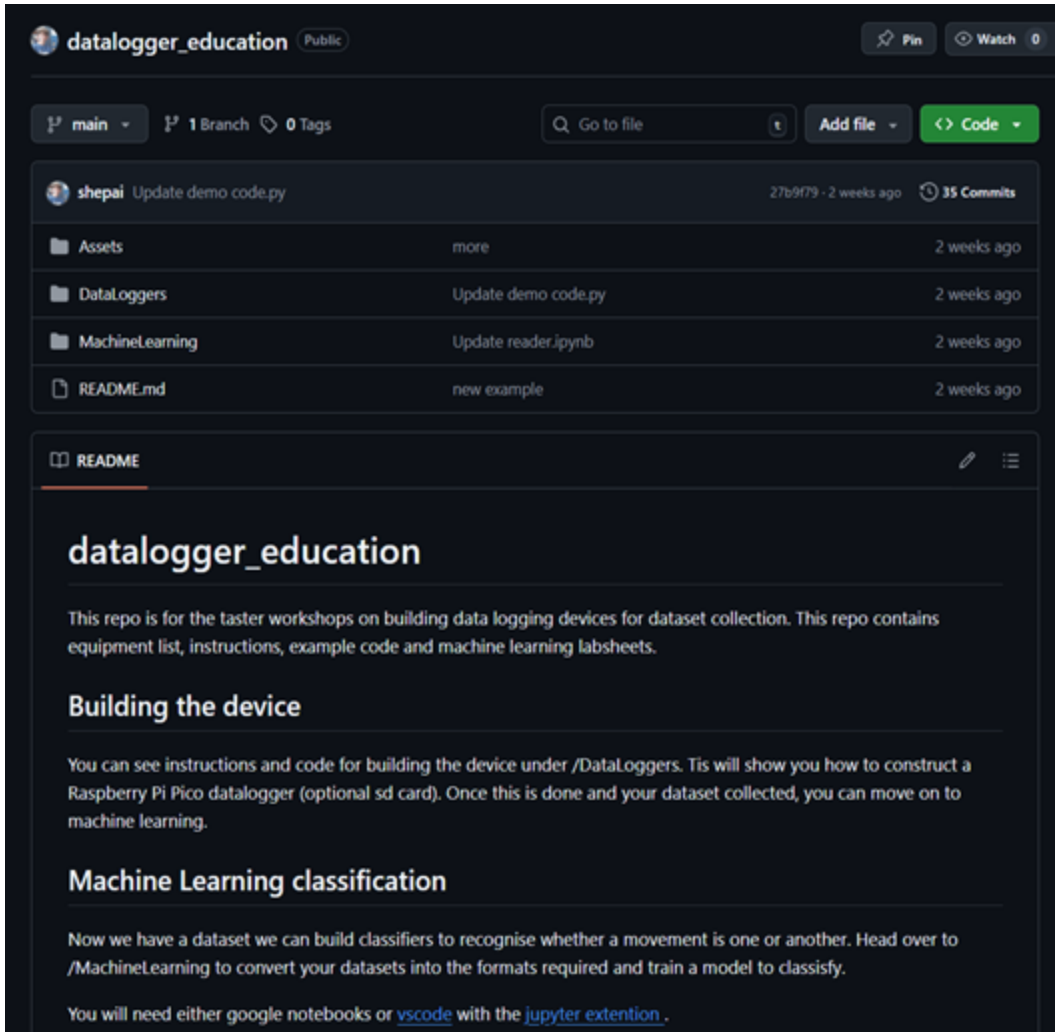


Today’s task

https://github.com/shepai/datalogger_education

Go to the website and follow the instructions for building the raspberry pi device

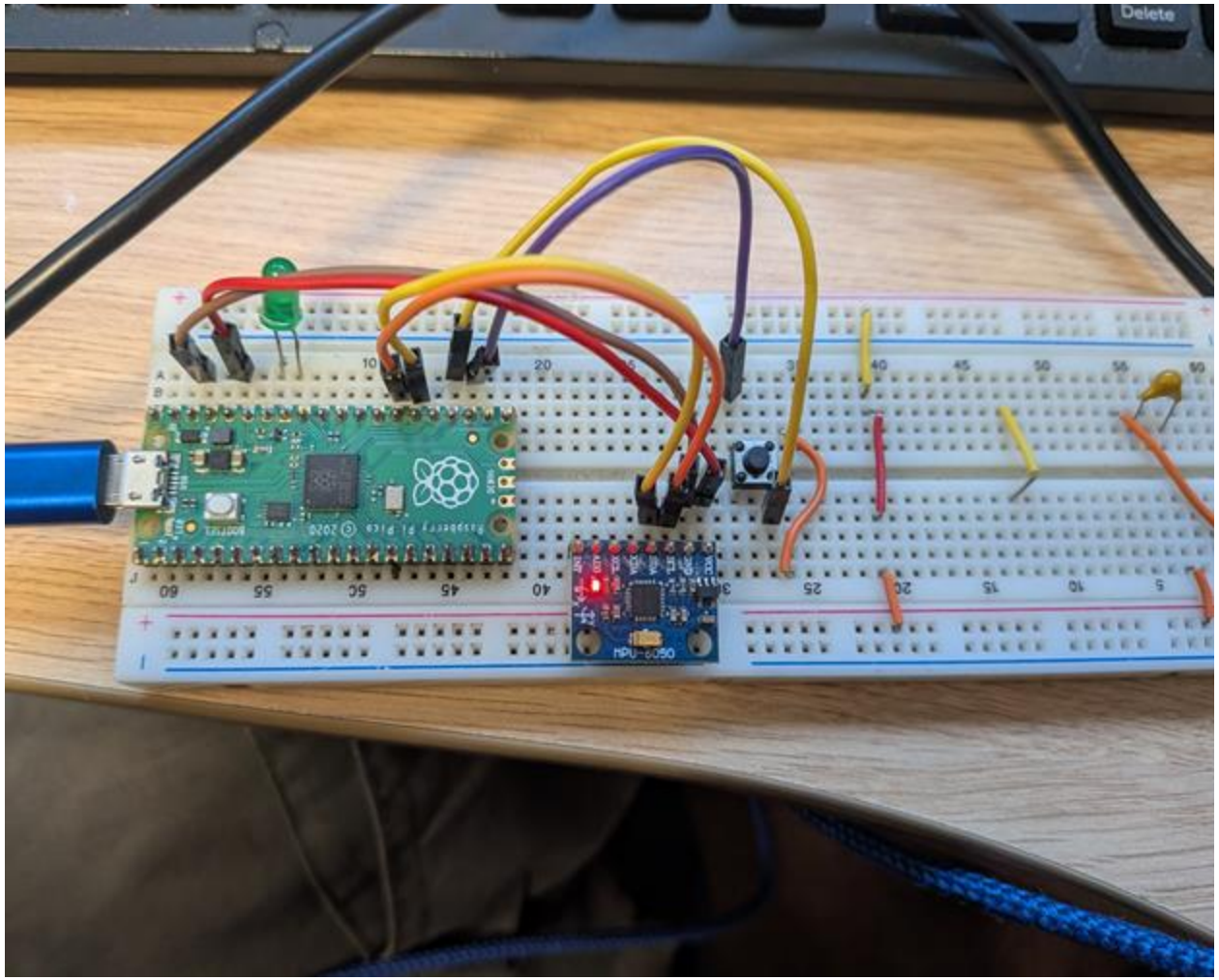
访问该网站，并按照说明构建 Raspberry Pi 设备。



End result

You will need to download Thonny IDE

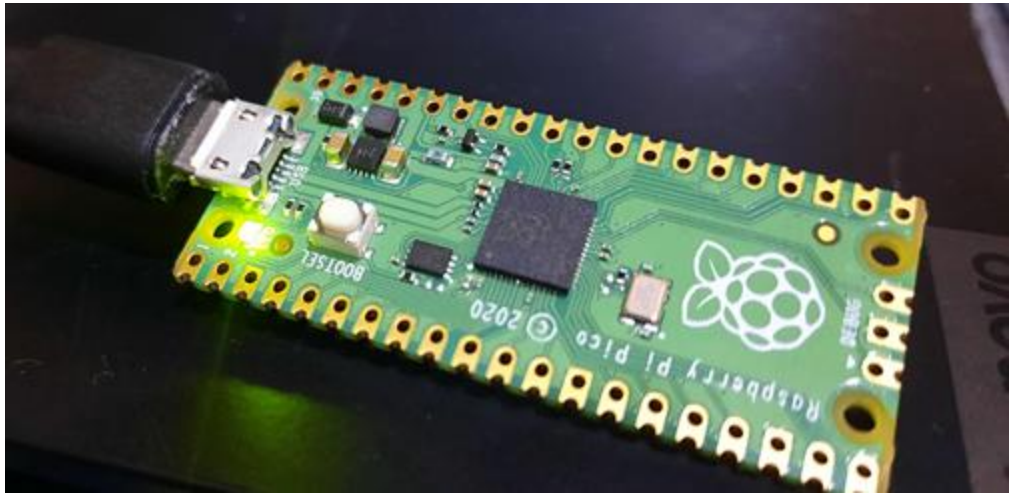
你需要下载 Thonny 集成开发环境 (IDE)

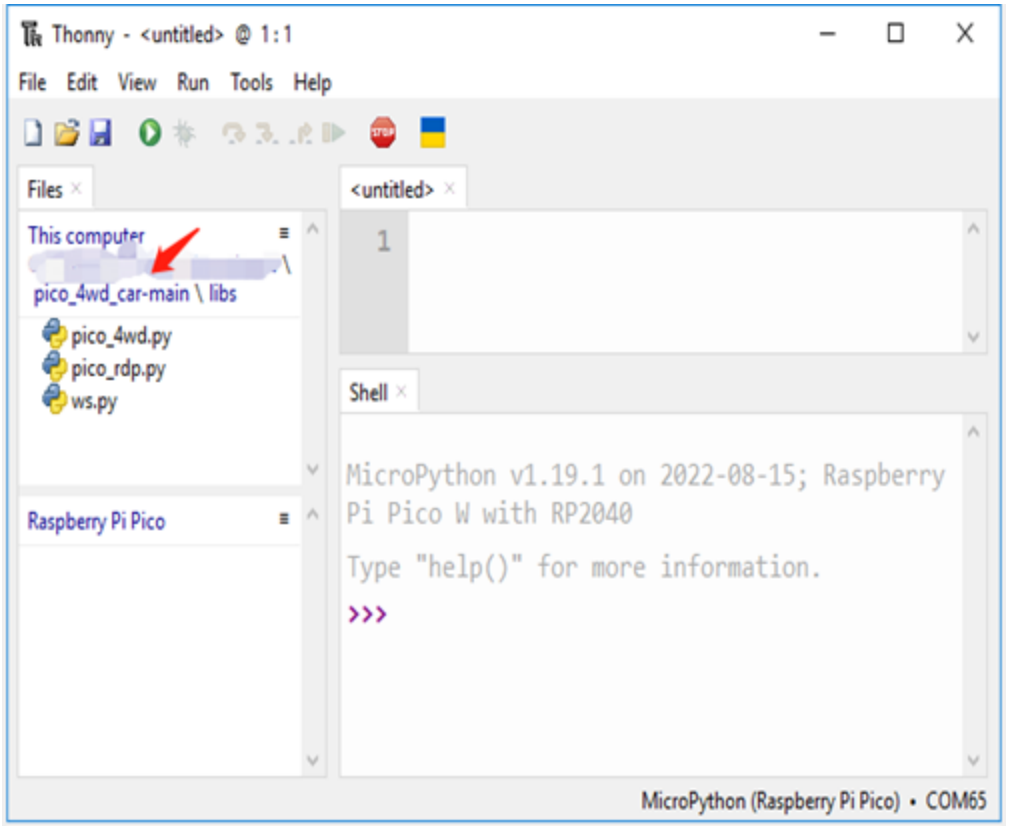


Uploading code

Plug it in via USB - 通过 USB 将它连接上。

Open Thonny IDE - 打开 Thonny 集成开发环境 (IDE)。

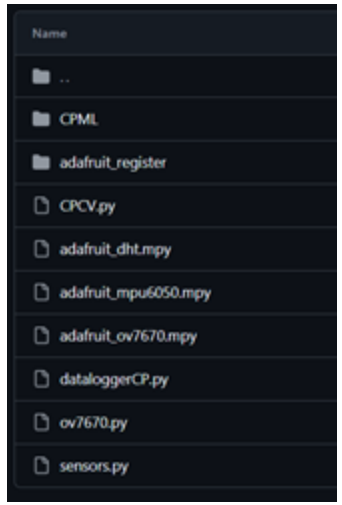




Uploading code

In the GitHub there is a folder called CircuitPython dependencies. Copy and paste the content into the /lib directory

在 GitHub 上有一个名为 “CircuitPython dependencies” 的文件夹。将其中的内容复制并粘贴到 /lib 目录中



Uploading code

Copy the contents over from demo code and paste it into code.py on the device

将 demo 代码中的内容复制过来，并粘贴到设备上的 code.py 文件中。

datalogger_education / DataLoggers / demo code.py

shepai Update demo code.py

Code Blame 45 lines (43 loc) · 1.16 KB

```
#import the data logging lib
import busio
import digitalio
import board
from dataloggerCP import datalogger
import adafruit_mpu6050
import time

#set up the pins
spi = busio.SPI(board.GP2, board.GP3, board.GP4)
cs = board.GP1
i2c = busio.I2C(board.GP21,board.GP20) #might need to flip it
btn=digitalio.DigitalInOut(board.GP18)
btn.direction=digitalio.Direction.INPUT
btn.pull=digitalio.Pull.UP
LED=digitalio.DigitalInOut(board.GP28)
LED.direction=digitalio.Direction.OUTPUT
#set up the sensor
mpu = adafruit_mpu6050.MPU6050(i2c)
logger=datalogger(spi,cs)
#read line function from the sensor
def read():
    xa,ya,za=mpu.acceleration
    xg,yg,zg=mpu.gyro
    return str(xa)+","+str(ya)+","+str(za)+","+str(xg)+","+str(yg)+","+str(zg)+"\n"
```

Gathering datasets

Think of at least two actions (for example running and walking)

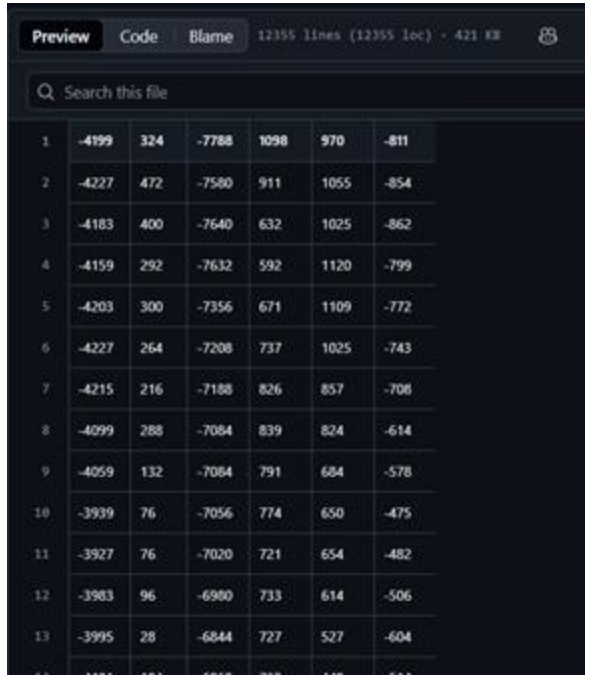
Press the button on your circuit to begin recording, and take note of what you did while recording data. Press the button again to stop recording.

Record a few actions and then back them up and label them on your computer

想出至少两个动作（例如跑步和走路）。

按下电路板上的按钮开始录制，并在录制数据时记下你做了什么动作。再次按下按钮停止录制。

录制几个动作，然后将数据备份到你的电脑上并进行标注。



1	-4199	324	-7788	1098	970	-811
2	-4227	472	-7580	911	1055	-854
3	-4183	400	-7640	632	1025	-862
4	-4159	292	-7632	592	1120	-799
5	-4203	300	-7356	671	1109	-772
6	-4227	264	-7208	737	1025	-743
7	-4215	216	-7188	826	857	-708
8	-4099	288	-7084	839	824	-614
9	-4059	132	-7084	791	684	-578
10	-3939	76	-7056	774	650	-475
11	-3927	76	-7020	721	654	-482
12	-3983	96	-6980	733	614	-506
13	-3995	28	-6844	727	527	-604

Machine learning

Once we have gathered the data sets we can make classifiers

I suggest downloading vscode for this task, and install the jupyter extension

We will be working from something called notebooks

一旦我们收集好了数据集，就可以构建分类器。

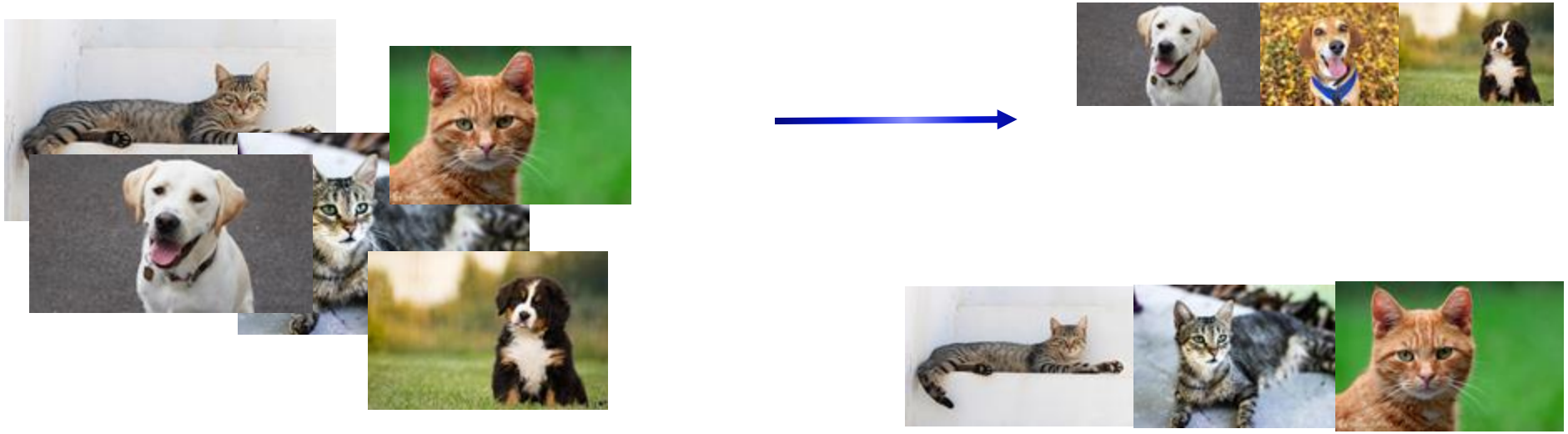
我建议为此下载 VS Code，并安装 Jupyter 扩展。

我们将使用一种叫做 notebooks（笔记本）的方式来进行工作。

What is a classifier?

an algorithm that automatically categorizes data into one or more predefined classes

一种能够将数据自动分类到一个或多个预定义类别中的算法。



How we will be using it

We will be using simple classifiers to classify our datasets. Work through the code on the Github.

How does your dataset do? Can you find ways to improve it?

Things to look into:

PCA and data augmentation

Different models

我们将使用简单的分类器来对我们的数据集进行分类。请按照 Github 上的代码一步步运行。你的数据集表现如何？你能找到改进的方法吗？

可以研究的内容：

- PCA 和数据增强
- 不同的模型

datalogger_education / MachineLearning /

Welcome to Sussex

Expand the World with US

UNIVERSITY OF SUSSEX